Build a Paper Prototype

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Build a Paper Prototype before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Design the Decision Process

Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design **Instruction**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Build a Paper Prototype trace and record the first step you can explain without help.

Use W2: Run and record the first match for Building and tracing a paper **prototype**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Build a Paper Prototype explanation against the supplied facts before discussing it.

Use the worked case for Building and tracing a paper **prototype**. Proposed explanation: “A correct-looking answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Build a new route through a small catalogue

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Build a Paper Prototype target to a trace and a justified conclusion.

Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Build a Paper Prototype case and show the deciding evidence.

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

> How can a learner simulate it?

Show your **trace**, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Your paper **prototype** shows an **input** card, a decision card and an **output** card. Explain what a person must simulate and what the **prototype** cannot prove.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 32 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Build a Paper Prototype adds to Helpful Classroom Sorter and keep the

> evidence.

Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Assemble the runnable kit

> Assemble the runnable kit. Apply the supplied Build a Paper Prototype facts and keep your reasoning.

Create P1 version 1 using the three **instruction** cards in the lesson. Copy catalogue B version 1 and make a request card, current-row marker and **trace** sheet. List what each item records.

Explain which information stays fixed during one run and what changes.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Run and record the first match

> Run and record the first match. Apply the supplied Build a Paper Prototype facts and keep your reasoning.

Use the manual checklist to predict P1's **output** for 11 minutes No. Then ask a partner to operate your saved cards. Record the **expected result** separately from the observed steps and **output**, including the reason.

Explain why the **operator** should stop without checking later rows.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Locate the source of a wrong result

> Locate the source of a wrong result. Apply the supplied Build a Paper Prototype facts and keep your reasoning.

Saved **instruction**s P0 say “start at B1; if B1 fails, report no match.” For the valid request 9 minutes No, an **operator** follows P0 exactly and reports no match. Identify the **instruction** defect and write the changed step.

Explain how to label the correction and what to retain from the failed run.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Build a new route through a small catalogue

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a new route through a small catalogue. Apply the supplied Build a Paper Prototype facts and keep your reasoning.

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Make the evidence repeatable

> Make the evidence repeatable. Apply the supplied Build a Paper Prototype facts and keep your reasoning.

A teammate keeps only a final “B2” answer and throws away the request and old **instruction**s after editing them. Explain why another person cannot check this result.

List what to save for each run and explain why an observed error must remain in the record.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Build a Paper Prototype

> A fictional team turns a storage-bin specification into movable instruction cards. Explain the labelled step before checking the changed condition.

A fictional team turns a storage-bin specification into movable **instruction** cards.

Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

AI ACADEMY / AI-1/32

Build a Paper Prototype

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Expected result

Baseline checks X then Y and expects Y for count 4.

2. Operate literally

Follow the current prototype cards without secretly repairing them.

3. Actual trace

Prototype stops at X; it never performs the capacity comparison.

4. Record mismatch

Expected Y, actual X: locate the missing check.

CHANGE ONE THING

Insert "if capacity is below count, continue" before accepting a bin.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What should appear in the test log before and after the repair?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional team turns a storage-bin specification into movable **instruction** cards.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Expected Y, actual X: locate the missing check.

1. Actual **trace**

2. Record mismatch

3. Operate literally

4. **Expected result**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Record mismatch Expected Y, actual X: locate the missing check.

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Insert “if capacity is below count, continue” before accepting a bin. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The revised **trace** rejects X and accepts Y for 4.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The revised **trace** rejects X

and accepts Y for 4. 3<4 is true; 7<4 is false, so Y fits.

Repair a missing building block

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Prompts Are Design **Instruction**s (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

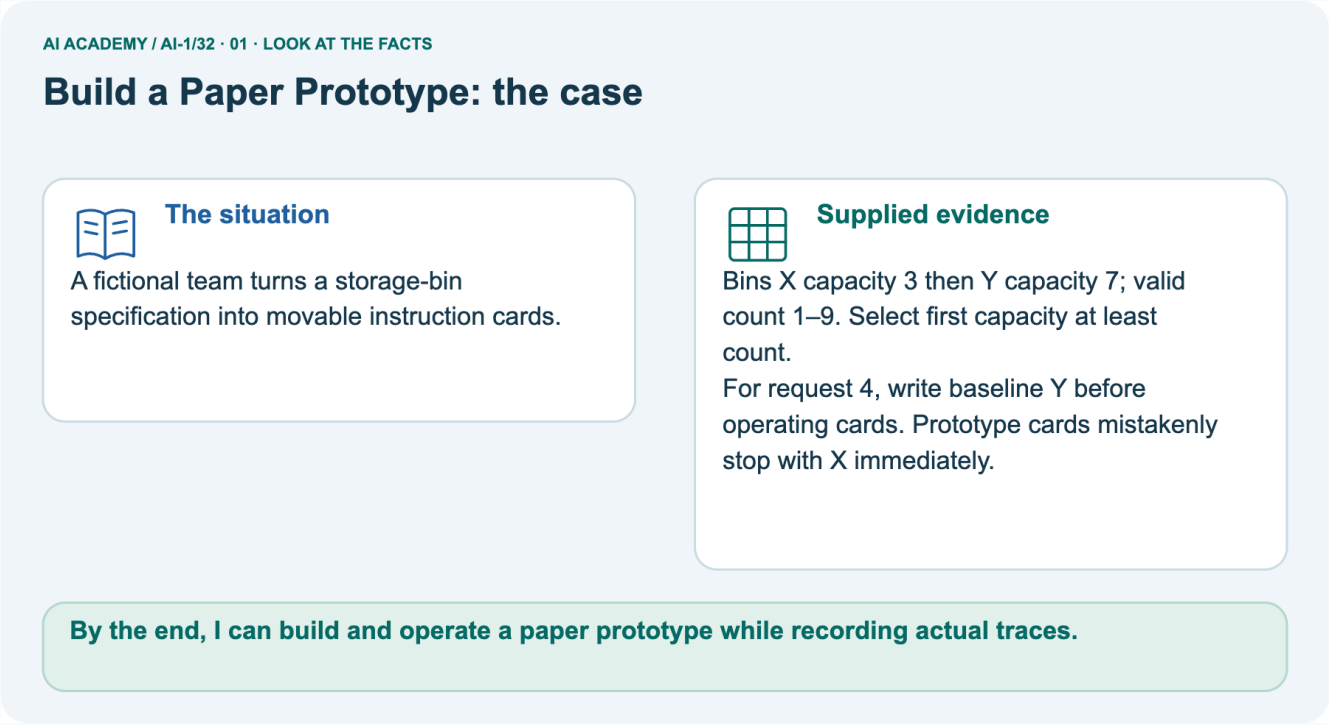
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Build a Paper Prototype picture from its labels, then record a relevant limit.

01. Build a Paper Prototype: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

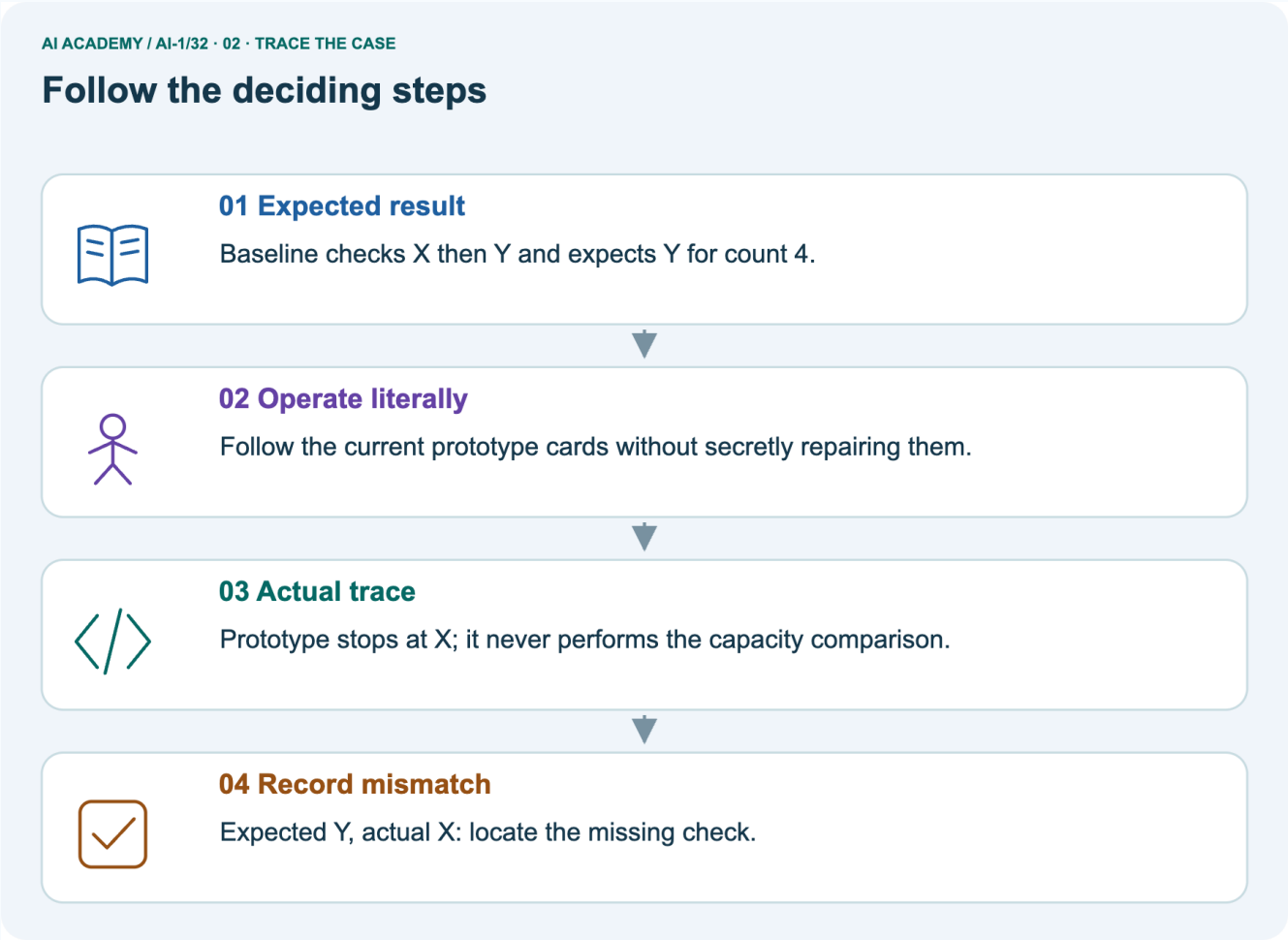
A fictional team turns a storage-bin specification into movable **instruction** cards. Supplied facts: Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



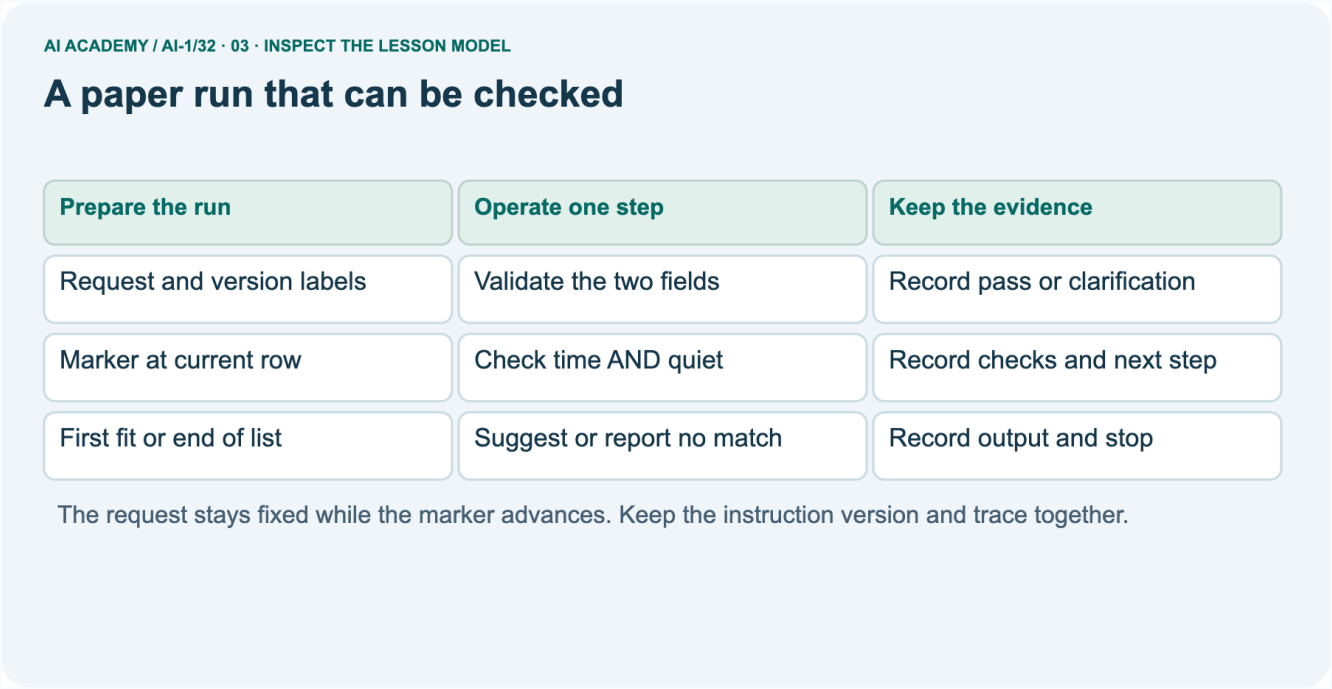
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected result: Baseline checks X then Y and expects Y for count 4. → Operate literally: Follow the current **prototype** cards without secretly repairing them. → Actual **trace**: **Prototype** stops at X; it never performs the capacity comparison. → Record mismatch: Expected Y, actual X: locate the missing check.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A paper run that can be checked



Follow the labelled picture with the matching explanation. Open the vector diagram

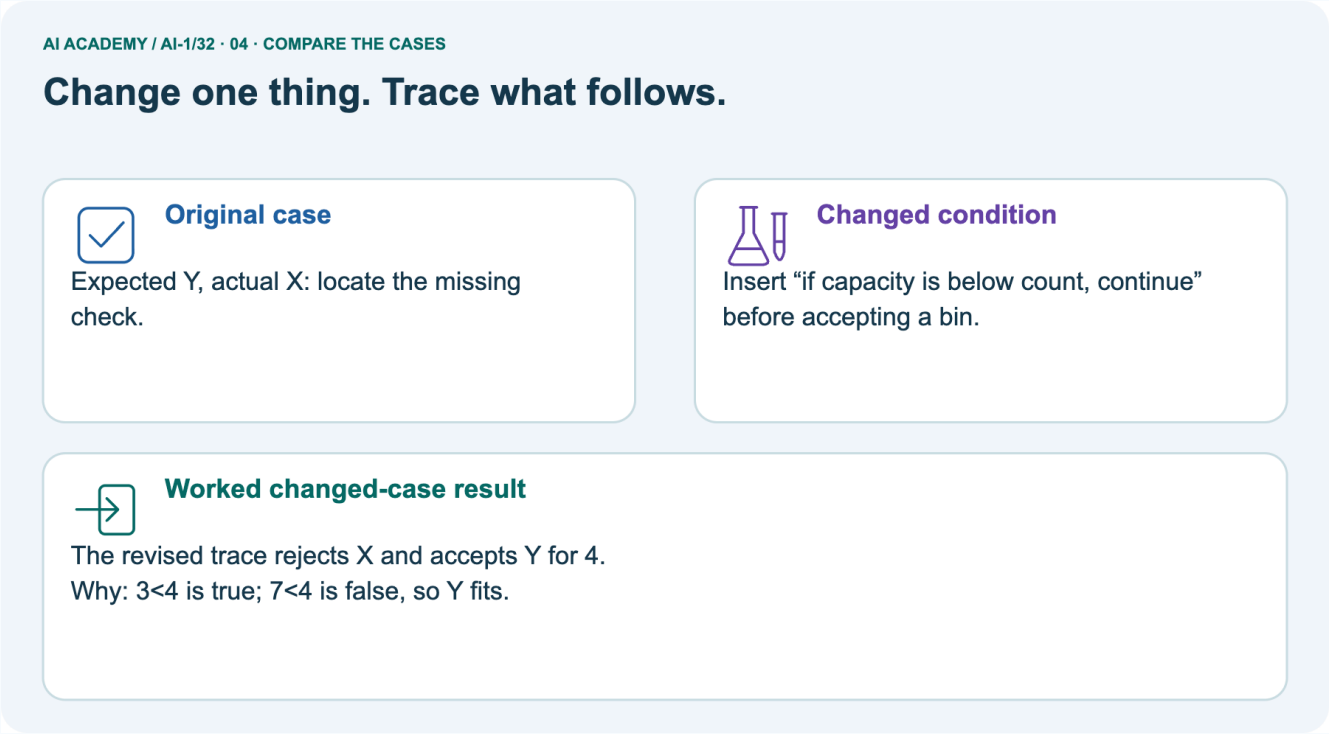
A paper run that can be checked. Request and version labels | Validate the two fields | Record pass or clarification. Marker at current row | Check time AND quiet | Record checks and next step.

First fit or end of list | Suggest or report no match | Record **output** and stop

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

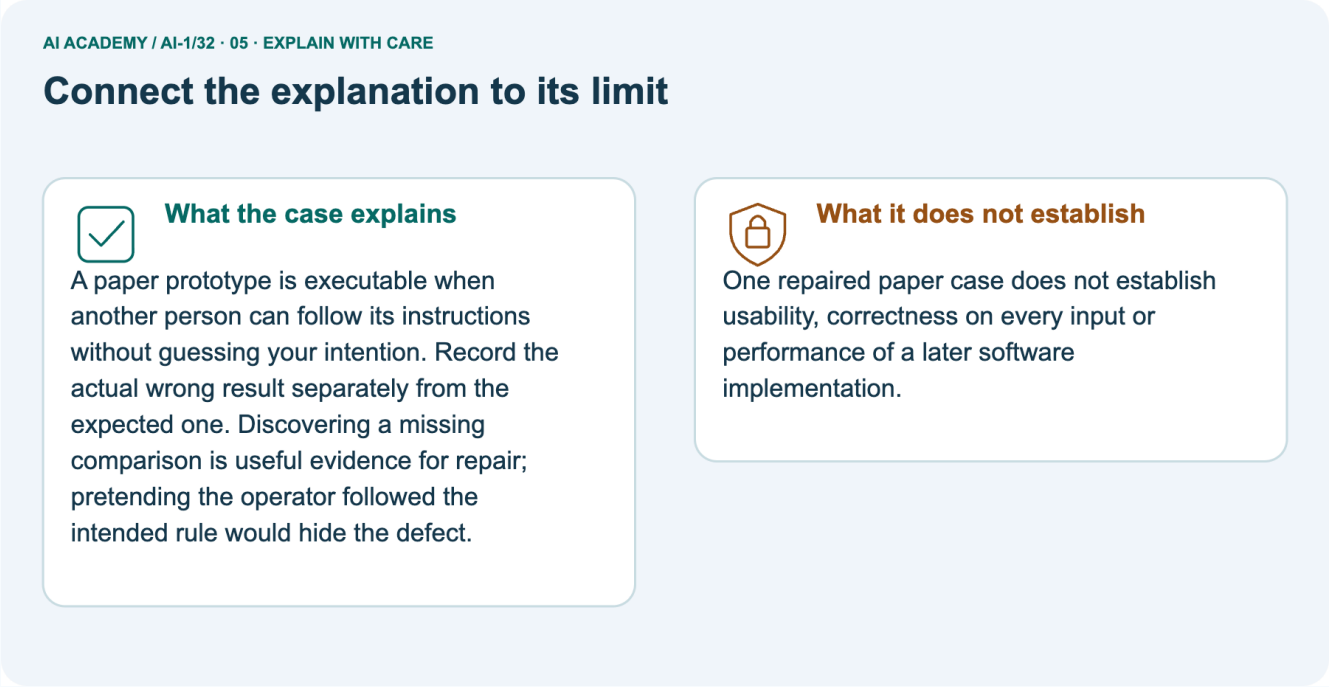
Original case: Expected Y, actual X: locate the missing check. Changed condition: Insert “if capacity is below count, continue” before accepting a bin. **Expected result**: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

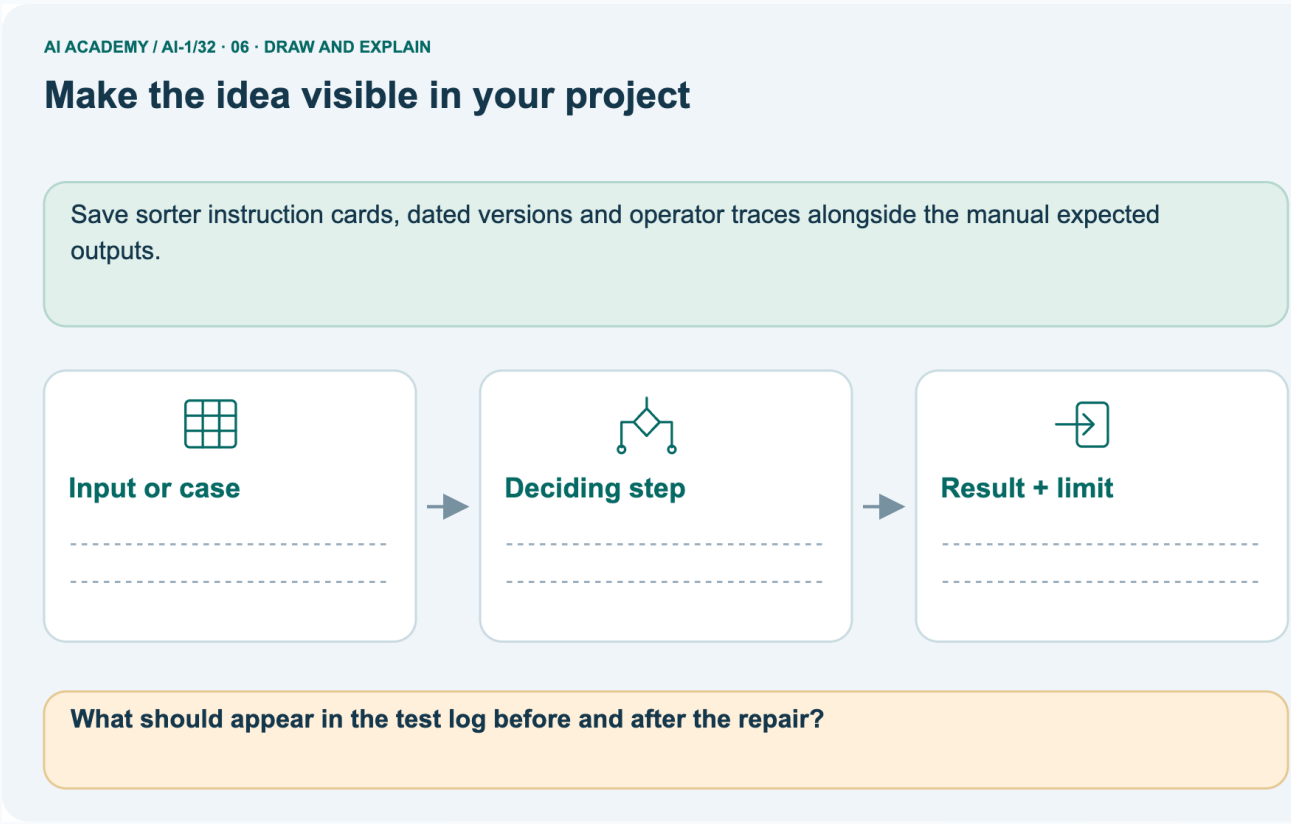
Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Limit: One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s. Draw your **input**, deciding step, result and limit.

Supported question: What should appear in the test log before and after the repair?

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.